INSPECTION COORDINATION PROCEDURE BETWEEN LTHE, TPIA AND VENDOR

FOR

LSTK PACKAGE-1

OF

"ETHYLENE GLYCOL" AT PARADIP REFINERY, ODISHA

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1. PURPOSE

The purpose of this document is to provide guidelines for the Inspection co-ordination among Vendors, Sub-Vendors, Third Party Inspection Agency (TPIA) and LTHE in LSTK Package-1 for "Ethylene Glycol" at Paradip Refinery, Odisha.

2. SCOPE

The scope of this procedure is to detail about the procedure and coordination between Vendor/Sub-vendor, LTHE, TPIA and PMC/Owner during Shop Quality Control and Inspection phase, for the equipment / items to be supplied for LSTK Package-1 for "Ethylene Glycol". The requirement of this procedure is applicable to all Vendors for the supply of equipment, packages and materials (E&I, Piping etc.).

3. DEFINITION

Term	Definitions
OWNER / CLIENT	**INDIAN OIL CORPORATION LIMITED, PARADIP or Their Authorized Agency**
PMC	**TOYO ENGINEERING INDIA PVT. LTD**
LSTK CONTRACTOR	**LTHE Hydrocarbon Engineering Limited (LTHE)** The Person(s), Firm, Company, Organization responsible for executing the specified work inclusive of Engineering, Procurement, Construction and Commissioning on lump sum turnkey basis.
LICENSOR	**EGU : Scientific Design** **ERU : Mc Dermott International**
VENDOR	The person(s), firm, company, organization from whom LTHE procures products / services as part of their obligation towards Owner for the project.
ITP / QCP / QAP	**Inspection and Test Plan / Quality Control Plan / Quality Assurance Plan** Vendor's ITP/QCP/QAP which includes various inspection activities and scope of Vendor, LTHE/TPIA, OWNER/PMC in terms of Review, Witness and Hold points during Inspection.
PIM	**Pre-Inspection Meeting** A Pre-Inspection Meeting / Pre-Fabrication Meeting will be held to ensure that technical requirements of the Purchase Requisitions are understood by Vendors & TPIA. During PIM, Vendor and TPIA (as required) representatives will be briefed for project requirements, material certification, traceability, PMI,

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	and LTHE/TPIA & PMC/Owner's involvement, to help minimize downstream problems.
Inspection Category	The level of intervention (QA/QC) to be applied by LTHE, Manufacturer or Vendor in order to assure the quality of the procured items taking into account the inspection categorization and indicative ITP.
Inspection Notice	The Project "Inspection Notice" Form will be used by Vendor / Sub-Vendor, to invite LTHE/TPIA & PMC/Owner/Appointed TPIA as per agreed inspection visits, through ICMS.
NCR	Non-Conformance with Project Specification and QMS
MRB / MDB / VDR	**Manufacturing Record Book / Manufacturing Data Book / Vendor Data Record** MRB/MDB/VDRs are reviewed by LTHE/TPIA Inspector during manufacturing and on completion. MRB/MDB/VDR include Quality Records in accordance with Project Document, which is attachment to the Requisition.
TPIA	LTHE appointed Approved Inspection Agency
Hold Point (H)	Hold Point is a critical step in manufacturing and testing where it is essential that LTHE/TPIA, PMC/Owner's representative (as applicable) inspects component / equipment / review record to ascertain that the product for delivery complies with specified requirements. Process activities designated as a 'Hold Point' shall not proceed unless LTHE/TPIA, PMC/Owner's representative is present or 'Hold Point' is formally waived and the waiver is confirmed in writing. Waiving of a 'Hold Point' shall be documented and the reason for the waiving stated.
Witness Point (W)	Witness point is a critical step in manufacturing and testing where it is desirable that LTHE/TPIA, PMC/Owner's representative (as applicable) inspects component / equipment to ascertain that the product for delivery complies with specified requirements. LTHE/TPIA, PMC/Owner's representative must be given requisite Inspection Notice for all process activities designated with 'Witness Point'. Once proper notification is given, the activitiy proceeds according to the schedule regardless of LTHE/TPIA, PMC/Owner representative's attendance, unless notified otherwise.
Review Point (R)	Review point is a stage designated in the ITP/QAP which requires the concerned agencies to verify the documents for their correctness and to confirm that the said documents meet the requirements laid down.

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Approval Point (A)	A critical step in design, procurement, and/or fabrication and testing, where it is mandatory that the LTHE/PMC/Owner approve a document / qualification / equipment or activity. LTHE/TPIA, PMC/Owner shall be notified at a duration specified in the contract, in advance of the first occurrence of the activity or provided the document in a period of time specified by the contract. The activity may not proceed without the LTHE/TPIA, PMC/Owner's approval.
Random Inspection (I)	Random Inspection of Activities as designated in the ITP/QAP. Vendor shall perform the activity after proper notification has been given for witnessing the activity. Vendor is not obliged to hold further processing, unless notified otherwise.
Monitor Point (M)	Critical step in manufacturing and testing where it is optional that LTHE/TPIA, PMC/Owner's representative (as applicable) inspects component / equipment in order to ascertain that the product for delivery complies with specified requirements. LTHE/TPIA, PMC/Owner shall be notified with requisite Inspection Notice, in advance of the first occurrence of the activity at each worksite. The activity may proceed with or without LTHE/TPIA, PMC/Owner's attendance.
Inspection	The process of measuring, examining, testing, gauging or otherwise comparing the items with the applicable requirements.

4. ABBREVIATIONS

AI	Authorized Inspector
AS	Alloy Steel
ASME	American Society Of Mechanical Engineers
ASNT	American Society For Non Destructive Testing
CIMFR	Central Institue of Mining and Fuel Research
CRA	Corrosion Resistance Alloy
CS	Carbon Steel
DSS	Duplex Stainless Steel
E&I	Electrical & Instrumentation
FAT	Factory Acceptance Test
FIR	Field Inspection Report
IBR	Indian Boiler Regulations
ICMS	Inspection Call Monitoring System

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IR	Inspection Report
IRN/IC	Inspection Release Note/ Inspection Certificate
ITP	Inspection And Test Plan
LTHE	L&T Hydrocarbon Engineering Limited
MDB	Manufacturing Data Book
MR	Material Requisition
MRB	Manufacturer's Record Book
MRT	Mechanical Run Test
MTC	Material Test Certificate
NCR	Nonconformance Report
NDE	Nondestructive Examination
NDT	Nondestructive testing
PESO	Petroleum and Explosives Safety Organisation
PIM	Pre Inspection Meeting
PMC	Project Management Consultant
PMI	Positive Material Identification
PO	Purchase Order
PQP	Project Quality Plan
PQR	Procedure Qualification Record
PR	Purchase Requisition
PS	Purchase Specifiction
PWHT	Post Weld Heat Treatment
QA/QC	Quality Assurance/Quality control
QAP/QCP	Quality Assurance Plan/ Quality Control Plan
QMS	Quality Management System
SS	Stainless Steel
TPI / TPIA	Third Party Inspector / Third Party Inspection Agency
VDR	Vendor Data Record
WPQ	Welder Performance Qualification
WPS	Welding Procedure Specification

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5. REFERENCE

a) IEGP-6351-8640-TP-000-001, Rev. T2- Inspection Methodology & Co-ordination Procedure for EPCC/LTHE or Package Supplier.
b) PDRP-8440-SP-0024, Rev. F3 – Equipment Surveillance
c) Special Conditions of Contract – Part-A
d) Project Quality Plan (Doc No: IPEG-6389-8830-QA-000-0001)
e) Inspection Categorization and Surveillance Plan (Doc No: IPEG-6389-8830-QA-000-0003)
f) Shop Inspection Methodology (Doc No: IPEG-6389-8830-QA-000-0002)

Note: - Latest revision of LTHE generated documents as referred above shall be followed.

6. GENERAL

6.1 All equipment, packages and materials shall be procured from PMC/Owner approved vendors furnished in the contract documents irrespective of the equipment and materials being procured by the Vendor / Sub Vendor / LTHE directly or through its contracted agencies / or as approved by PMC/Owner in due course of project.

6.2 L&T and / or TPIA and / or PMC and / or IOCL shall have the right to visit Vendor's / Sub-Vendor's works / Independent Test Houses for the purpose of verification or auditing, any of the inspection functions as per the approved ITP / QAP. Vendor shall be bound to take necessary corrective actions as desired by them.

6.3 It is the responsibility of the Vendor / Sub-Vendor for the entire inspection co-ordination with TPIA / PMC as applicable.

6.4 Wrapping / Coating and Painting material shall also be procured from manufacturers approved by Owner / PMC.

6.5 Some of the critical items/ licensed items shall have to be procured from vendors approved by Process licensors (if applicable).

6.6 Project Approved Third Party Inspection Agencies (TPIA) for performing inspection activities on behalf of LTHE are as given below. For imported Equipment / Materials, group companies of below listed TPIA shall be utilized for inspection.

I. Projects and Development India Ltd.
II. TUV SUD South Asia Pvt. Ltd.
III. IR Class Systems and Solutions Private Limited.
IV. VCS Quality Services India Pvt. Ltd.
V. Certification Engineers International Limited.
VI. DNV GL Business Assurance India Pvt. Ltd.
VII. Tata Projects Ltd.
VIII. Apave Assessments India Pvt. Ltd. (Previously Known as Vincotte International India Assessment Services Pvt. Ltd.)
IX. Lloyds Register Asia
X. Bureau Veritas

6.7 Items / Equipment have been categorized, for the purpose of inspection coverage, into three categories, namely Category A, B & C based on Surveillance grade (as applicable). Inspection

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coverage shall be as per extent of surveillance required as per Surveillance grade, Inspection categorization and Indicative ITPs. The list of items / equipment with assigned Inspection Category / Surveillance grade are mentioned in "Inspection Categorization and Surveillance Plan (Doc No.: IPEG-6389-8830-QA-000-0003)".

- CATEGORY A : ITP/QAP approval by LTHE, TPIA & PMC/Owner. Stage and final inspection including document review as per approved ITP/QAP by TPIA. Final inspection including document review by PMC/Owner.
- CATEGORY B : ITP/QAP approval by LTHE, TPIA & PMC/Owner. Stage and final inspection including document review as per approved ITP/QAP by TPIA. Document review by PMC/Owner.
- CATEGORY C : ITP/QAP approval by LTHE, TPIA & PMC/Owner. Verification and/or stage and/or final inspection including document review by TPIA. No inspection by PMC/Owner.

Vendor/Sub-Vendor shall conduct all inspection activities in addition to the contents of this Inspection Categories, as per its own Quality Control Program, Code, Standards & OWNER/PMC Specification requirements.

6.8 **For Coded Equipment (U/U2):** Inspection shall be carried out by AI as well as TPIA. Authorized Inspectors (AI) shall be appointed by Vendor. However appointment of TPIA shall be in LTHE's scope

6.9 **Statutory Inspection Requirements:** Vendor to ensure and comply statutory inspection and certification requirements for IBR, PESO etc. as applicable per data sheet. Vendor to submit certificates issued by respective authority for LTHE/TPIA verification and records.

6.10 **For IBR Material / Equipment**:

Equipment / Material that come under the purview of IBR shall be inspected by IBR or IBR approved inspection agency.

Vendor to ensure compliance to all statutory requirements as per Indian Boiler Regulations (IBR), for IBR material / equipment. All materials, fabrication, welding, testing etc. shall meet the requirements of IBR. All IBR approved design calculations and drawings shall be submitted to LTHE/PMC/Owner for records.

Vendor to submit IBR certificate in original for supply. In case of non-availability of original certificate / part supply, photocopy of the original certificate duly attested by the local boiler inspector where the vendor is located is the minimum requirement for acceptance of material.

6.11 **For Electrical and Instrument Item :** All electrical equipment & instrument, which are suitable for installation in hazardous area as specified in the datasheet, shall have valid PESO approval along with applicable ATEX / IEC-Ex / CIMFR or equivalent agency certificate. Indigenous E&I items shall be supplied with valid BIS license. PESO approval is mandatory for items procured within India & from outside.

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6.12 For following items, inspection by LTHE/TPIA and/or PMC/Owner is not applicable. PMC/Owner shall accept these materials based on review of manufacturer's TC / Guarantee Certificates. Relevant Certificates shall be submitted at site for PMC/Owner review and acceptance.

- Cement
- Reinforced bars
- Structural Steel material, Gratings & IS2062 plates sourced from manufacturers
- Hand railing material
- Wrapping & Coating material
- Paints
- Chemicals and Cleaning agents
- Oil lubricants
- Insulation material
- Earthing material
- Hardware used at site for Electrical & Instrument items

7. REVIEW CODES

Approval comments on the submitted documents shall carry one of the following codes:

STATUS CODE	DESCRIPTION	EXPLANATION
1	Rejected	Revise and Resubmit. Release to Manufacture must be given in Writing by LTHE / PMC / Owner
2	Comments as noted	Revise and Resubmit. Work may proceed subject to incorporation of comments.
3	Acceptable	Incorporate comments, if any. Resubmit as Certified Final. Work may proceed.
4	No Comments	Acceptable. Certified Final Issue. Do Not resubmit. Work may proceed.
5	Information Only	Accepted for Information only. Do not resubmit. Work may proceed.

8. INSPECTION AND TEST PLANS / QAP

8.1 Inspection and test plans attached to the requisition are generic and indicative only. Vendor shall submit a detailed Inspection & Test Plan, based on Indicative ITP's, Codes, Standards, categorization of items, applicable drawings / data sheets, Material Specification, purchase requisitions, PMC/owner Specification, Statutory requirements (if any) & lesson learnt for review and approval by LTHE, TPIA & PMC/Owner as applicable.

8.2 Vendor shall prepare the ITP / QAP / QCP as per the sample format attached (Refer Exhibit I for Typical ITP format) and forward to LTHE. ITP / QAP / QCP format of Vendor/ Sub-Vendor

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may be acceptable if all the contents of Exhibit I are covered and is mutually agreed by LTHE and Vendor.

8.3 Each ITP shall identify the inspection category and detail all witness, hold and certification review points as specified in the requisition, referenced specification and standards.

8.4 Vendor's ITP shall identify and include the following as a minimum.

a) A document sequence of Inspection and Test activities related to the product or equipment being supplied.
b) Details of specific procedure, method and work instruction to be applied to each activity shall be defined.
c) The extent of examination and acceptance criteria for each activity shall be defined.
d) Method of verification of sub vendor's conformance to requirement.
e) Incoming, in – process and final inspection requirements.
f) Calibration of measuring & test equipment's and maintenance of calibration records.
g) Statistical techniques, if employed to be defined.
h) The specific allocation of responsibilities for each activity, including identification of inspection authority responsible for the release of conforming products
i) Traceability requirements as applicable

8.5 Vendor shall include all statutory inspection requirements in the Inspection and Test Plan of all stamped equipment (ASME, IBR, CCE etc.).

8.6 Approval of ITP/QAP in at least Code-2 (Comments as Noted) by LTHE, TPIA & PMC/ Owner (as applicable) is mandatory prior to commencement of manufacturing activities.

8.7 For Category 'A', 'B' & 'C' items Inspection & Test Plans shall be submitted to PMC/Owner, after review / approval by LTHE & TPIA. For this purpose, LTHE shall arrange the access of the following documents.

a) Un-priced Purchase Order / Letter of Intent
b) Approved Purchase Requisition
c) Applicable drawings / data sheets
d) Agreed Deviations / Variations, if any.

8.8 For the Inspection Category 'A' and 'B' Static and Rotary items, agreed/approved ITPs shall be held by the vendor and initialed progressively by the vendor's inspector, LTHE/TPIA, PMC/Owner and any other authorized inspectors (as applicable), to demonstrate successful completion of each inspection activity.

8.9 For Packages, package vendor shall submit list of sub-items covered in the package along with category of inspection, for LTHE & PMC/Owner approval (as required). Based on the Inspection category, Package Vendor shall submit detailed Inspection &Test Plans, for major sub-ordered items within package, for approval by LTHE, TPIA and PMC/Owner.

8.10 Package vendor shall submit following documents to LTHE for approval of Sub-vendor ITP.

a) Un-priced Sub-vendor Purchase Order / Letter of Intent
b) Relevant portion of approved PS, applicable drawings / data sheets.
c) Agreed Deviations / Variations

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This ITP/QAP/QCP shall be reviewed and signed by Sub-Vendor & Vendor before submitting to LTHE & TPIA for review & approval. LTHE will obtain PMC approval on this ITP/QAP/QCP as per project requirement.

In case of multiple ITP/QCP/QAPs for package items, Vendor/Sub-vendor need to provide cover page listing out all ITP/QCP details with document number, title and revisions.

8.11 It is Vendor's prime responsibility to update and maintain ITP/QAP/QCP and other QA/QC procedures as a live document and shall revise and re-submit for approval of LTHE and PMC/Owner as and when significant changes are identified in MR, Drawing, Data Sheet etc. and subsequent approval shall be obtained from LTHE, TPIA and PMC/Owner.

8.12 For Piping Bulks (Pipes, Fittings, Flanges, Gaskets and Fasteners), item specific standard ITP/QAP shall be prepared by LTHE considering project Requirements. The standard ITP/QAP shall be submitted to PMC/Owner for approval. The Approved ITP/QAP shall be sent to vendor while preorder / post order stage. Vendor/Sub-Vendor shall study the ITP/QAP requirements and confirm their understanding and compliance to ITP/QAP & to the PMC/Owner project specific requirements. Vendor's singed & endorsed ITP/QAP shall be forwarded to LTHE well in advance before start the manufacturing activities.

9. IDENTIFICATION AND TRACEABILITY

9.1 Identification and traceability of materials, component and equipment shall be maintained and documented throughout the pre-fabrication and installation process for all pressure retaining components and non-pressure retaining components welded to pressure retaining components including welding consumables, paints, coatings, adhesives and CRA internals. Additional precautions shall be applied for manufacturing SS / DSS / CRA and low temperature materials. Preferably, low stress stamps should be used for identification stamping. In case of paint/permanent markers are used for marking on SS material, the same shall be Chloride free.

9.2 All equipment including spares shall be identified, tagged, marked, colour coded as stated in either the requisition or other referenced documents. Paint and Markers shall not contain harmful metals or metal salts such as zinc, aluminium, lead or chlorides, Sulphur or other halogens that may be harmful to the alloy.

9.3 Vendor shall maintain traceable control of all materials listed in 9.1 above, and shall keep a record of their certified chemical and mechanical properties (as applicable).

10. PROCESS CONTROL

10.1 All welding and related activities (e.g. Post Weld Heat Treatment and Non Destructive Examination) shall satisfy the requirements of relevant international standards & Project Specific requirements as specified in the contract / purchase order.

10.2 Special processes (e.g. Welding, Painting, PWHT, NDE, Coating, Cable jointing etc.) shall be carried out under supervision of a competent person as defined by the respective specifications. All special processes shall be executed in accordance with documented and qualified procedures that ensure work is complying with OWNER/PMC technical specifications.

10.3 The Vendor shall maintain an index or register of all approved welders and NDE operator continuity records which shall be made available to LTHE/TPIA/PMC/Owner when requested.

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The register shall give information on training of the welders and date and results of qualification tests. WPS number, base metal, type of welding consumable, joint design, welding position etc. shall also be recorded in the register.

10.4 Process control documents as required by the ITP/QAP/QCP shall be submitted well in advance prior to commencement of the activities for review and approval (as applicable and as stated in the ITP/QAP/QCP). They shall be available at the location where the activities are carried out.

11. PIM (PRE-INSPECTION MEETING)

11.1 PIM shall be organized with Vendors &/or major sub vendors before commencement of manufacturing activities for Inspection category 'A' & 'B' tagged items covered under Surveillance Grade 1, to confirm PMC/Owner quality requirements.

11.2 After Placement of Purchase orders, Vendors / Sub-Vendors shall initiate the Pre-Inspection Meeting (PIM) through Inspection Call Monitoring System (ICMS).

11.3 Vendor and Major Sub-Vendor, LTHE's representative including Quality personnel, TPIA, shall participate in PIM. The basic purpose of PIM is to discuss the vendor scope, purchase order quality requirements in detail and to ensure that these requirements are fully understood by all parties including TPIA. In addition, Witness and Hold points shall be discussed.

11.4 Vendor/Sub-vendor shall ensure that Engineering & QA/QC documents necessary for PIM are submitted to LTHE minimum 30 days advance of schedule PIM.

11.5 The minimum documents which are required to be approved in Code-2 (Comments as Noted) from LTHE prior to Pre-inspection Meeting are as below:

- Material / Purchase Requisition
- Data sheet & GA Drawing
- ITP / QCP
- WPS / PQR (As applicable)

11.6 Vendor shall give LTHE/TPIA/PMC/Owner at least 10 (ten) working days notification prior to PIM date (in case of vendor works within India) and at least 30 (thirty) working days notification prior to PIM date (in case of vendor works outside India), through ICMS.

11.7 During PIM, following points will be discussed and agreed between Vendor/Sub-Vendor, TPIA & LTHE.

- The scope of shop inspection and tests, availability of applicable project documents, such as purchase orders, requisitions, specifications and Vendor's drawings and procedures.
- Vendor's shop organization QA/QC organogram for the project will be explained by Vendor.
- Coordination procedures, such as communication channels among the parties concerned and submission of a Weekly / monthly inspection schedule (For Next two Months), applications, records, etc. will be discussed and confirmed at the meeting.
- Communication protocol for QA/QC documents submission / approval.
- Vendor's quality control system to be demonstrated. LTHE will consolidate the information furnished during the pre-award stage.

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- Status of applicable documents viz. Specification, ITP/QAP, WPS / PQR, Inspection & Test procedures.
- Contract requirements to ensure Vendor's and TPIAs understanding of its responsibilities.
- Sequence of manufacturing, testing will be explained by Vendor.
- Inspection and Test Plan / related procedures will be reviewed and finally approved by the LTHE, TPIA & PMC/Owner (as applicable).
- Inspection items to be witnessed by the LTHE/TPIA and PMC/Owner and also other matters not confirmed yet, such as hold points and detailed inspection measures, will be taken up and decided.
- PMC/Owner special requirements if any.
- NDE requirements, Hold Points & Witness Points.
- Inspection notification methodology.
- Discussion for Sub-Vendor requirements.
- Manufacturing / Inspection Schedule.
- Vendor Quality Control Program check, inspection hold points.

12. INSPECTION CO-ORDINATION

12.1 Vendor shall forward inspection notification to LTHE & TPIA coordinator through ICMS only, as per agreed communication protocol prior to the scheduled inspection for LTHE / TPIA for participation in inspection and/or test as per approved ITP/QAP.

Vendor shall necessarily fill up all the required details in ICMS before sending the inspection calls to LTHE and TPIA.

12.2 The minimum Inspection notification period for inspection activities for Hold, Witness, Review, Surveillance etc. (as required) as per approved ITP/QAP shall be as given below:

Inspection Parties ↓	**Notification Period**			
	Within India (Notice Days)	**Within India (Re-Schedule / Cancellation Notice Days)**	**Outside India (Notice Days)**	**Outside India (Re-Schedule / Cancellation Notice Days)**
LTHE appointed TPIA	3 Working Days	2 Working Days	7 Working Days	2 Working Days
PMC / Owner	10 Working Days (2 Weeks)	2 Working Days	30 Working Days (6 Weeks)	4 Working Days

12.3 Wherever, more than one inspection agency is involved for single inspection activity, inspection by all the agencies shall be done concurrently to avoid duplication.

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12.4 Any inspection activities which call for PMC/Owner inspection as per approved ITP / QAP, Vendor shall clearly mention the PMC/Owner intervention in the Inspection Notice and forward to LTHE Discipline QC Engineer, through ICMS.

12.5 Vendor / Sub-Vendor shall plan the inspection visits required by LTHE/TPIA in such a manner so as to achieve maximum inspection stages attended with minimum possible inspection visits.

12.6 In case any delay is experienced in availing the TPIA inspection based on inspection calls and as per approved ITP/QCP, immediate information shall be furnished to LTHE's Discipline QC Engineer with details, for taking up with TPIA.

12.7 At the end of each month, Vendor shall provide 2 (Two) month look ahead inspection schedule to LTHE Discipline QC Engineer, for TPIA participations.

13. MANUFACTURING AND INSPECTION ACTIVITIES AT VENDOR'S SHOP

13.1 Manufacturing and stage wise inspection can proceed with Code-2 (Comments as Noted) approved documents like ITP/QCP, applicable procedures, drawings & data sheets. However, Final Inspection shall be carried out based on drawings and documents approved by LTHE/TPIA, PMC/Owner (as applicable), in Code-3 (Acceptable) / Code-4 (No Comments) / Code-5 (For Information). Vendor to ensure approval of drawings and documents accordingly, before raising Inspection Notice to LTHE/TPIA.

13.2 It is the responsibility of the Vendor to furnish copies of the latest revision of the approved specifications, drawings, documents, ITP/QAP & Procedures to LTHE/TPIA /OWNER/PMC and ensure that the inspection is carried out as per the latest revisions and approved documents.

13.3 In addition, vendor shall be responsible for ensuring availability of all tools and equipment (duly calibrated) that may be required for satisfaction of an inspection / test.

13.4 Vendor shall submit all applicable procedures including NDT, PWHT, WPS/PQR, Weld & NDE Plan etc., to LTHE/TPIA for review and approval, as per the approved ITP / QAP. For these documents, approval by PMC/Owner is not applicable. However, for category 'A' static items WPS / PQR with weld & test plan and NDT plan approved by LTHE/TPIA shall be submitted to PMC/Owner for review.

13.5 For items/equipment, which involve multiple weld joints and welding procedures, Vendor/Sub-vendor shall prepare detailed Weld Map/Plan and submit it along with WPS and PQR for approval by LTHE/TPIA.

13.6 Advanced NDT methods, automatic NDT techniques and cladding applications procedure may requires LTHEs approval. Vendor shall submit these special procedures well in advance for necessary approvals.

13.7 WPS/PQR already approved by reputed inspection agencies may be used by the vendor if found meeting the requirements and these shall be reviewed and accepted by the LTHE/TPIA. Similarly, welder qualification record witnessed and approved by the reputed inspection agencies and meeting the requirements of applicable project specifications and design code shall be reviewed and accepted by TPIA at the time of inspection.

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13.8 Specialized operators shall be qualified and certified NDT operators employed, if any, shall be certified as per ASNT/ISNT and shall have valid certificates.

13.9 LTHE/TPIA shall ensure that all personnel engaged in performing the activities have requisite qualification and relevant work experience, wherever required.

13.10 For the Vendor/Sub-vendor, all the stages of inspection to be performed by them, shall be HOLD point and no stage shall be offered to LTHE/TPIA, PMC/Owner without having performed/witnessed such stages internally.

13.11 Old stock / degraded / deteriorated / earlier used materials shall not be used for the project..

13.12 Piping material such as pipes, fittings, flanges, gaskets, fasteners etc., shall be colour coded by the vendor to avoid possibilities of mix-up at site. Details of colour code shall be as specified in engineering specifications and/or procedure. Vendor and/or TPI shall check the colour coding requirements during the final inspections / prior to dispatch of materials.

13.13 Piping materials shall be stamped / identified by the manufacturers as per the material specifications. This is to ensure availability of the markings (viz., heat number, material specification etc.) after welding with other materials at site during traceability verification. It is mandatory to check these markings and location by the Vendor and /or TPIA before dispatch, to ensure compliance to Code & Material Specification requirements.

13.14 TPIA shall carry out hard / Low Stress stamping / Tagging / stencilling for Identification, Traceability and Verification at site. Details for the same shall be mentioned in TPIA IRN/ IC.

13.15 All materials and welds of Alloy steel & SS shall be PMI tested under witness by TPIA (as per approved ITP / QAP), by an Alloy analyser as per the PMI Specification (DOC NO: IEGP-6351-8640-TP-000-002). LTHE shall prepare and submit project specific PMI Procedure to PMC / Owner for review and approval. PMC/Owner approved project specific PMI procedure shall be forwarded to Vendor, for implementation.

13.16 Vendors / Sub-Vendors shall check the Ferrite content of all the welds in SS component / item. The Ferrite content shall be as per approved PR/PS/Design code.

13.17 Vendor shall progressively offer documents / reports for TPIA review, as and when inspection visits are scheduled, and not at the end i.e. completion of all manufacturing activities.

13.18 Vendor / Sub-vendor shall ensure that there is no deviation / non-conformity with respect to approved PR, Applicable specifications, Codes etc. However, in case any deviation / non-conformity is observed, the same shall be immediately reported to LTHE, by the Vendor / Sub-vendor / TPIA, for necessary resolution.

13.19 Vendor shall take special care to protect the final painting of the equipment / items during handling, storage, loading etc., in order to minimize touch up and re-work at site.

13.20 Vendor shall also comply with the procedure for Preservation and Packing of Equipment and Materials for shipment.

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14. WITNESS INSPECTION AND TESTS & INSPECTION CLEARANCE

14.1 Stage wise inspection can proceed with Code-2 (Comments as Noted) approved documents like ITP/QCP, applicable procedures, drawings & data sheets.

14.2 TPIA shall ensure that final inspection is carried out based on drawings & documents reviewed by LTHE in Code-3 (Acceptable) / Code-4 (No Comments) / Code-5 (For Information) and specify accordingly in Final inspection Certificate / Inspection Release Note along with the list of documents / drawings and their revision number.

14.3 LTHE/TPIA, PMC/ Owner/ Process Licensor shall have the right to visit Vendor's / Sub-Vendor's works / Independent Test houses for the purpose of verification or auditing any of the inspection functions in addition to those as per the approved ITP / QAP. Vendor shall be responsible to take necessary corrective actions as desired by them if any item is found to be not meeting specification requirements.

14.4 TPIA shall issue Field Inspection Report (FIR) / Inspection Report (IR) after each visit. The same shall be uploaded in ICMS immediately upon completion of inspections.

14.5 In case, inspection cannot be completed or undertaken due to reasons such as non-readiness of material, back up documents etc., such incidents and reasons shall be recorded in FIR / IR and jointly signed by the Vendor and the TPIA. The same shall be mailed to LTHE from Vendor's works.

14.6 Before judging final acceptance of the equipment and materials at the time of final inspection, vendor's final documents shall be reviewed by the TPIA to verify the following:

- All specified inspections and testing have been completed.
- Results of inspections and testing conform to specified requirements.
- Documents have been prepared to prove that the finished product meets the specified requirements.

14.7 Inspection / Surveillance of work by LTHE / TPIA is intended to verify the effectiveness of the vendor's own inspection or test activities and such supplementary inspection shall not relieve the vendors of their obligations with regard to conformity of the work to PMC/Owner Specifications, Codes and Standards requirements. The category of LTHE/TPIA's inspection / surveillance may change during the course of the work, depending on Vendors performance.

14.8 TPIA shall ensure and declare compliance to approved ITP by endorsing the approved ITP and making it part of the Final Inspection Certificate / Inspection Release Note, for all Category 'A' & 'B' Static and Rotary items. No material shall be shipped with open NCR.

14.9 LTHE/TPIA reserves the right to verify the compilation of documentation during final inspection and / or prior to hydro test for all items. Vendor shall necessarily co-operate during such verification and comply with the specified requirements

14.10 Vendor shall forward Inspection Release Note / Certificate issued by TPIA along with following reports (as minimum) endorsed by TPIA & PMC/Owner (as applicable) to LTHE Discipline QC Engineer. Without written clearance from LTHE, items shall not be dispatched to site.

- **Static item**: - IRN & IC of TPIA, Signed QAP/ITP endorsed for each stage by Vendor+TPIA, Hydro-test reports, Final dimensional Reports, Painting Reports, PMI reports, and MDR in

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case of Coded equipment, IBR forms-II, III, IV (alternatively, provisional certificate issued of IBR in absence of aforementioned forms) for IBR equipment.

- **Machinery / Rotary Item**: - IRN & IC of TPIA, Signed QAP/ITP endorsed for each stage by Vendor+TPIA, Painting Reports, PMI reports, Performance Test, MRT, Unit test Reports (As applicable).
- **E&I Item**: IRN & IC of TPIA, Function Test Report (Manufacturer's Test Certificate / Calibration Certificate), Statutory Certificates, Painting Reports, PMI reports (As applicable).
- **Piping and bulk**: IRN & IC of TPIA, Manufacturer TC's, Heat Treatment charts, Final dimension reports, PMI report, IBR forms for IBR items
- **Piping Spool**: IRN & IC of TPIA, History Sheet, Painting Reports, Hydro-test report (As applicable) PMI reports.
- **Fabricated Structure**: IRN & IC of TPIA, Painting Reports, History Sheet/Inspection Summary.
- **Packages**: Applicable above items and / or IRN & IC of TPIA, Final Dimensional Reports, Painting Reports, Functional test report, Statutory Certificates and other test reports / certificates as applicable.

Note: Vendor shall also submit specific additional documents if requested by LTHE for clearance.

14.11 LTHE will submit soft copy of TPIA IRN / IC for all items and TPIA endorsed ITP (for Category 'A' & 'B" Static and Rotary items) to PMC/Owner for information, at the time of dispatch.

14.12 PMC/Owner-Site shall clear Material Receipt Inspection Report (MRIR) based on photocopy / scan copy of IC / IRN issued by TPIA. Original documents shall be furnished for the clearance in case any deviation is observed.

15. TPIA DOCUMENTS

TPIA shall ensure that the Inspection Release Note / Certificate issued by TPIA shall mention the following details as a minimum:

- Project Name: LSTK Package - 1 for Ethylene Glycol at Paradip Refinery, Odisha
- Owner: Indian Oil Corporation Limited (IOCL)
- PMC: Toyo Engineering India Private Limited (TEIPL)
- LSTK Contractor: LTHE Hydrocarbon Engineering Limited (LTHE).
- LTHE's PO No. Including Amendments if any and Date
- Manufacturer's name
- Date & Place of Inspection
- Sub order No (vendor order reference for sub-ordered items) - if applicable
- Approved PR / MR Specification No. (including Rev. No. and Date)
- Documentation approval status i.e. Approved Drawing No. (if applicable) including Approval, Rev. No. and Date

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- Approved Inspection & Test Plan / QAP No.
- Item / Equipment description with TAG No (If applicable) as per PO.
- A list of MTC Nos for Bulk items
- Details of stages witnessed and documents reviewed as minimum as per approved ITP/QAP. i.e. Detail reference to Inspection Test Plan stage
- Quantity as ordered inspected and released.
- Status of the order whether complete or incomplete
- Problems/areas of concern if any
- Deviations, shortages if any
- New and outstanding non-conformances if any
- Certification on IR / IRN as **"INSPECTION HAS BEEN CARRIED OUT BASED ON FINAL APPROVED DRAWING & ITP."**
- Inspector name, stamp and signature.
- TPIA to issue separate release note (in detail) for mandatory and E & C spares indicating item description, Qty., etc.

16. SUPPLIER NON-CONFORMANCE

16.1 Vendor / TPIA shall inform LTHE of any non-conformance to the approved or stated requirements detected during Fabrication and/or detected by Inspector during his witness inspection within 24 hours of noticing. A non- conformance shall be notified to PMC/Owner in stipulated time.

16.2 If a non-conformance is found, Vendor shall immediately segregate the equipment or materials. Vendor shall carry out the required/necessary actions for the rectification, modification or discard, etc., in accordance with the agreed disposition plan.

16.3 The Project Manager jointly with Project Quality Manager shall classify those NCRs as High / Medium / Low risk based on following:

- High Risk : If significant effect on HSE or technical integrity of production facilities (immediate action required)
- Medium Risk: If significant effect on integrity of production facilities (and negligible effect on HSE (Action required within 7 working days max.)
- Low Risk: Conditions other than high risk and Medium risk (Action required within 15 working days but before close out of project)

16.4 Vendor need to perform root cause analysis and revert to the originator within 72 hours of issuance of NCR.

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16.5 Project Quality Manager in consultation with Project Manager shall take remedial action to correct non-conformance. The complete management of Non-conformance shall be done as per flow chart provided in attached EXHIBIT II.

17. FINAL DOCUMENTATION

17.1 Vendor shall submit Index of final documentation for LTHE review and approval. This index shall be in line with project approved format.

17.2 After approval of Index, Vendor shall submit soft copies of final documents as per approved Index, for review and acceptance of LTHE.

17.3 After acceptance of final documents (soft copy), Vendor shall submit number of hard copies of final documents as per the approved PR/MR & PO terms and conditions.

17.4 MTCs / Test reports and relevant TPI IRN / IC for spare items shall be separately compiled by Vendor and to be submitted along with the final documentation, as per approved Index.

18. LIST OF ATTACHMENTS

Exhibit I : Typical Inspection & Test Plan Format
Exhibit II : Flow Chart for NCR Management

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EXHIBIT I – TYPICAL INSPECTION AND TEST PLAN FORMAT

INSPECTION AND TEST PLAN

Purchaser	L&T Hydrocarbon Engineering	Item Description	
Project	LSTK Package-1for EthyleneGlycol at Paradip Refinery, Odisha	Item Number / Tag No.	
Client	Indian Oil Corporation (IOCL)	ITP Document No. & Rev.	
PMC	Toyo Engineering India Pvt. Ltd. (TEIPL)	MR / PR / Specification No.	
LTHE Job No.		Inspection Category	
Vendor		Name of TPIA	

Sl. No.	Inspection Activity	Extent of Check	Reference Document	Acceptance Criteria	Document required	Inspection Agency				Remark
						Sub-Vendor	Vendor	LTHE / TPIA	PMC / Owner *	

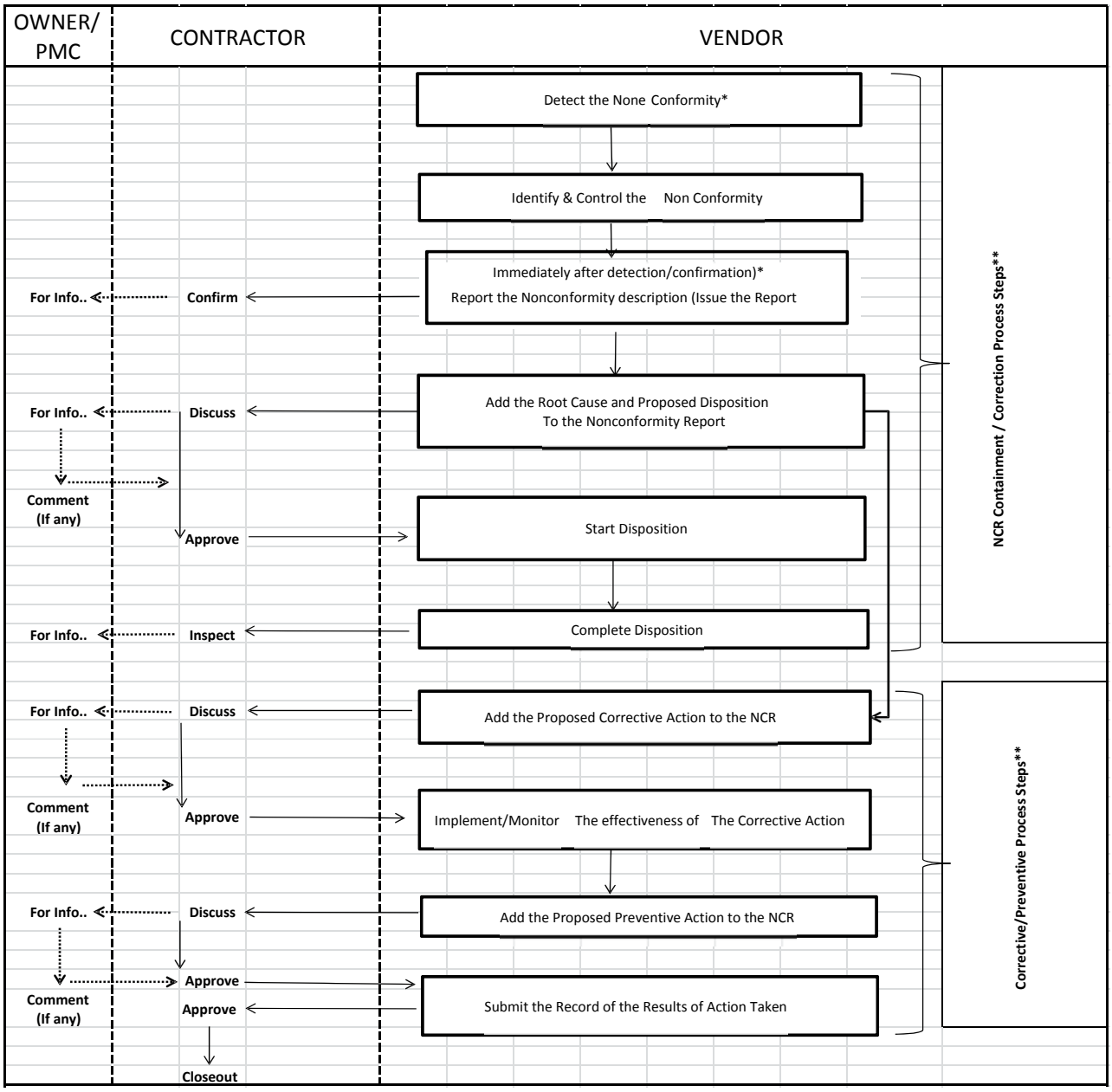
SPECIMEN

Legend : H - Hold ; W - Witness ; R - Review ; M - Monitor (For description please refer "Definition")
* Wherever applicable only
** For Category 'A' , 'B' & 'C' items

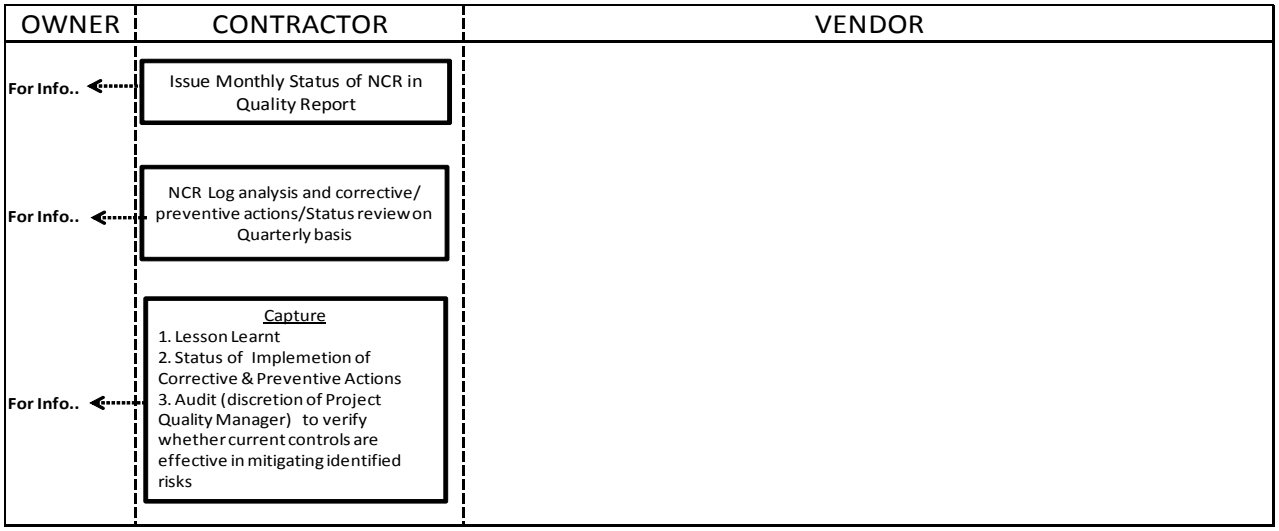
PREPARED & APPROVED BY (VENDOR)	APPROVED BY (LTHE)	APPROVED BY (TPIA)	APPROVED BY** (TEIPL)

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EXHIBIT II – FLOW CHART FOR NCR MANAGEMENT



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a) * Detection of Non Conformity & Reporting of Non conformity shall be done either by Vendor or Contractor appointed Inspection Agency or Owner Appointed Inspection Agency or Owner

b) **NCR Catagorization:

1. High Risk - Follow NCR Containment/Correction Process Steps & then Corrective/Preventive Process Steps.
2. Medium Risk - Follow NCR Containment/Correction Process Steps. Corrective/Preventive Process Steps under discretion of Project Quality Manager
3. Low Risk - Follow NCR Containment/Correction Process Steps. Corrective/Preventive Process Steps under discretion of Project Quality Manager

c) ** For Medium or Low risk NCR, Project Quality Manager may decide Corrective/Preventive action based on Repetative Items being ordered on the same Vendor/ reoccurance of similar NCR , thereby eleminating chances of Potential Non conformity

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